

The Optimizer Undoes It: Input-Affine Conditioning of a Selective Scan Is Gauge-Absorbable

A Training-Dynamics Null with a Mechanistic Explanation

Anonymous authors

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Abstract

Selective state-space models such as Mamba have reshaped long-context sequence modeling, yet it remains unclear which forms of conditioning actually persist under their training dynamics. We study FiLM-style regime conditioning of a Mamba backbone: a per-block channel-wise affine, which is the kernel-friendly, drop-in way to inject a regime signal. We find a decisive null. Under joint training the modulation collapses to identity, and it cannot be forced active even by direct supervision. We explain this mechanistically. The input affine’s scale is a *gauge direction* that folds losslessly into that block’s own $\Delta/\mathbf{B}/\mathbf{C}$ projections, so the optimizer slides the modulation back to identity. The decay shares a time constant ($\tau \approx 7000$ steps, $R^2 > 0.999$) across equities and crypto spot books, two structurally different markets. The null is separately confirmed on prediction markets. An output-side placement makes the null *overdetermined*. There the scale folds into the block’s bias-free output projection, while the lone non-absorbable shift decays in lockstep under reconstruction dominance. Escaping both would require modulating the scan parameters inside the fused kernel. A toy reference scan confirms that such an in-kernel modulation survives training, as does a char-level Mamba language model on a strongly decodable regime, which scopes the collapse to weakly-decodable drop-in conditioning. The mechanism is architecture-general by the algebra of the fold, though the training collapse is demonstrated only at our scale. The financial data is a demonstration substrate: under a matched-parameter ablation no backbone wins, and the one robust economic signal is matched by a logistic regression on the same features.

1 Introduction

We wanted a sequence model to behave differently in different regimes, so we looked for tools to condition the model on an inferred regime. For a Mamba backbone, the most kernel-friendly way to do this is a FiLM-style affine on each block’s input. The selective scan’s parameters $\Delta_t, \mathbf{B}_t, \mathbf{C}_t$ are themselves input-dependent (Section 3), so modifying the input alone lets regime information propagate into the scan without rewriting the fused CUDA kernel. We find that this implementation does not survive training. We create a zero-initialized FiLM modulator consisting of a regime router and a hypernetwork that emits per-block γ and β . The modulator is attached to a Mamba-3 world model and trained jointly with the host objective. The FiLM parameters are initialized to the identity transformation, making the modulated and unmodulated models bit-identical at initialization. To encourage the use of the modulation, we apply activation pressure through forced-active initialization and direct supervision of the router on a regime axis. Despite this, the modulation decays monotonically back to identity on every objective and dataset we test. The failure arises from training dynamics, not from a lack of expressive capability. On volatility-based regimes on FI-2010, a separately trained frozen router achieves 0.92 agreement with the regime buckets, demonstrating that meaningful regime information is present in the data. Yet, the FiLM parameters still converge back to identity during joint training.

The explanation is a gauge symmetry of the block parameterization. An input-side channel-wise scale folds into the block’s input projections as a right-multiplied diagonal $W_{\bullet} \text{diag}(\gamma)$, with no change to the function the block computes. These projections are bias-free up to a lone dt bias that can absorb only the shift’s Δ -component, so the shift as a whole cannot fold into the host and instead decays as an unrewarded direction (Section 5). Identity is therefore a gauge direction: the optimizer slides along it, returning (γ, β) to $(1, 0)$. This matches our observation that the decay signature is a property of the block, not the data. A forced-active escalation decays at the same rate ($\tau \approx 7000$ steps, $R^2 > 0.999$) on FI-2010 and Kaggle crypto, two different markets trained with the same optimizer.

We then run the logical follow-up to the gauge argument: an output-side placement that gates the block’s output. It does not escape absorption either. Each Mamba-3 block ends with a bias-free output projection, so the output scale is gauge-absorbed there with the same decay constant. The remaining additive shift decays in lockstep under reconstruction dominance. This leaves the in-kernel modulation as the only escape from the null.

Why this is a foundation-model question. Although we instantiate the study on a financial world model, the object of study is the Mamba block itself: gauge absorption is an algebraic property of the block’s parameterization. A channel-wise affine on the input folds into the same $W_{\Delta}/W_B/W_C$ projections whether the block sits in a world model, a language model, or any other selective-scan foundation model. The mechanism and its design consequence are therefore architecture-general by construction: algebraic and scale-free in principle. The training collapse it predicts, however, is demonstrated only at our scale, not yet at FM scale. The char-level Mamba language model control (Section 5) plays a narrow role: locating the boundary where FiLM starts working. The universality claim itself rests on the algebraic gauge argument.

Contributions.

1. A decisive, mechanistically explained null for input-affine FiLM conditioning of a selective scan. We claim the modulation collapses to identity under joint training and cannot be forced active, across five objectives and three LOB distributions (Section 4).
2. The mechanism behind the null: *gauge absorption*, distilled into a structural *absorbability criterion* (Section 5) that predicts which placements survive optimization. It carries a dataset-invariant signature ($\tau \approx 7000$, $R^2 > 0.999$), and an output-side placement makes the null overdetermined, locating the boundary inside the kernel. We demonstrate this rather than merely argue it, via a toy reference-scan experiment and a no-weight-decay ablation that shows the decay is weight-decay-driven over a flat loss (Section 5).
3. A boundary at regime decodability: a Mamba LM control (prose vs. code) shows a strongly-decodable regime engages the modulation, scoping the collapse to weakly-decodable conditioning (Section 5).
4. A load-bearing-diagnostic hazard (the gap reads positive while the mechanism is inert) and a demonstration substrate with an architecture-agnosticism check and a deflated, architecture-independent economic coda (Section 6).

The financial machinery (a Mamba-3 world model in the DreamerV3/Drama lineage) is the stress-substrate that let us probe the mechanism on genuinely different data. We describe it only to the depth the null requires.

2 Related Work

Selective state-space models and Mamba. S4, S5, and Mamba give $O(n)$ -complexity alternatives to attention (Gu et al., 2022; Smith et al., 2023; Gu and Dao, 2024). Mamba (Gu and Dao, 2024) introduced selective state spaces in which Δ and B are input-dependent, enabling content-aware filtering; Mamba-2 (Dao and Gu, 2024) cast this as structured masked attention; Mamba-3 (Lahoti et al., 2026) adds MIMO state transitions. (Although we used Mamba-3, we were unable to take advantage of MIMO due to hardware constraints.)

93 Our object of study is not a new SSM but how a selective scan trains under an added affine.
 94 The very input-dependence which makes the modulation seem natural is what makes it
 95 gauge-absorbable.

96 **FiLM and conditioning.** Feature-wise linear modulation (Perez et al., 2018) injects context
 97 efficiently via per-channel affine parameters. It is a default conditioning primitive across
 98 vision and multi-task learning. We are not aware of prior work that asks whether a FiLM
 99 affine persists under joint training of a selective scan, or that identifies its absorption into
 100 the host projections as a gauge direction.

101 **Scale redundancy and scale-invariance.** It is a known theme in normalization literature
 102 that a channel-wise affine placed before a linear map is redundant. van Laarhoven (2017)
 103 shows that weights feeding a normalization layer are scale-invariant. Thus L2 regularization
 104 acts on their scale, and transitively on the effective learning rate, rather than as a magnitude
 105 control. As Hoffer et al. (2018) illustrates, $BN(\|w\|_2 \hat{w}x) = BN(\hat{w}x)$, where $BN(\cdot)$ denotes
 106 Batch Normalization, w is the weight vector, $\|w\|_2$ its Euclidean norm, $\hat{w} = w/\|w\|_2$ the
 107 normalized weight direction, and x the layer input.

108 Our null extends this invariance and redundancy to selective state-space models, where the
 109 absorbed affine transformation is not a static normalization gain but an input-dependent
 110 FiLM modulation acting through the scan projections. What is new is threefold: the input-
 111 dependent selective-scan instantiation; the dataset-invariant decay signature that attributes
 112 the collapse to the optimizer; and the output-side placement showing the null is overdeter-
 113 mined.

114 **World models and the LOB substrate.** DreamerV3 (Hafner et al., 2023) established the
 115 categorical-latent RSSM template, and Drama (Wang et al., 2025) replaced its recurrence
 116 with a Mamba-2 selective scan. We instantiate this lineage with a Mamba-3 backbone purely
 117 as a substrate: no spine result depends on the world-model framing beyond a reconstruction
 118 objective and three distinct distributions. The rest is substrate context. DeepLOB (Zhang et
 119 al., 2019) and the FI-2010 benchmark (Ntakaris et al., 2018) anchor LOB direction prediction;
 120 the LOBCAST benchmark (Prata et al., 2023) documents the in-distribution macro-F1 range,
 121 confirming our data carries signal; and classical regime-switching (Hamilton, 1989) and
 122 changepoint detection (Adams and MacKay, 2007) motivate a regime axis. We use the
 123 Kaufman efficiency-ratio predictability axis as the most defensible regime signal in the
 124 study.

125 3 Background

126 **Limit order books, for a non-finance reader.** A *limit order book* (LOB) is the live ledger
 127 an exchange keeps for an asset. At each instant it records how much volume rests on the
 128 buy and sell sides at every price near the current mid-price, and it updates many times a
 129 second as orders arrive, cancel, and execute. The task is to forecast the next short-horizon
 130 move, price direction, volatility, or order flow, from the evolving shape of this ladder. We
 131 deliberately span three kinds of book that pose *structurally different* learning problems, so
 132 that a single shared null across them is informative rather than a quirk of one dataset.
 133 **FI-2010** is regulated equities (Nasdaq Nordic): a centralized, fixed-hours, discrete-tick
 134 event stream with mature, comparatively stationary microstructure. **Crypto** spot books
 135 (BTC/ETH/ADA) trade 24/7 with no circuit breakers across fragmented venues, which
 136 gives heavier tails, sharper volatility clustering, and thinner, faster-moving books. We
 137 sample them at three resolutions (1 s/1 min/5 min), each a different signal-to-noise regime.
 138 **Polymarket** is not a continuous price at all but a binary prediction market. Each contract
 139 settles to 0 or 1 at a known expiry, so its price lives in $[0, 1]$ as a probability. That probability
 140 is driven by a time-to-expiry clock and the reference move since open, and the terminal
 141 label is near-deterministic. Equity microstructure, crypto volatility, and a bounded settling
 142 probability are about as far apart as sequence-forecasting distributions get. That is why the
 143 same conditioning collapsing identically on all three is what the null rests on.

Selective state-space model. A discretized SSM maps an input x_t to an output y_t through a latent state h_t , $h_t = \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t$, $y_t = \mathbf{C}h_t$, where $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$ are the discretized state-transition and input matrices and $\mathbf{C}$ is the output matrix. Mamba makes the scan parameters input-dependent via learned linear projections of the block input x_t :

$$\Delta_t = \sigma(W_\Delta x_t), \quad \mathbf{B}_t = W_B x_t, \quad \mathbf{C}_t = W_C x_t, \quad (1)$$

where W_Δ, W_B, W_C are learned projection matrices, σ is the softplus nonlinearity, Δ_t is the per-step discretization size, and $\mathbf{B}_t, \mathbf{C}_t$ are the now input-dependent scan parameters. This selection mechanism is the property a regime-conditioner would exploit: any affine transformation of x_t propagates into all three scan parameters. It is also, as Section 5 shows, the property that makes an input affine absorbable.

RegimeFiLM modulator. The modulator works in two stages. A regime head reads the stem output and infers a soft categorical distribution over R regimes, from which it reads out a low-dimensional regime embedding r_t ($R = 4$ on the FI-2010/Kaggle spot books, $R = 8$ on Polymarket). A hypernetwork then maps r_t to per-block FiLM parameters, bounded for bf16 stability:

$$\alpha_t = \text{softmax}(W_r x_t^{(0)}) \in \mathbb{R}^R, \quad r_t = \alpha_t^\top E, \quad \gamma^{(\ell)} = 1 + \tanh(\tilde{\gamma}^{(\ell)}), \quad \beta^{(\ell)} = \tanh(\tilde{\beta}^{(\ell)}), \quad (2)$$

where α_t is the soft regime distribution the router weights W_r produce from the stem output $x_t^{(0)}$, E is a learned regime-embedding table (its rows the per-regime embeddings), r_t is the resulting soft-averaged regime embedding, ℓ indexes the block, $\gamma^{(\ell)}$ and $\beta^{(\ell)}$ are that block's FiLM scale and shift, and $\tilde{\gamma}^{(\ell)}, \tilde{\beta}^{(\ell)}$ are the hypernetwork's raw (pre-tanh) outputs. The tanh bounds $\gamma^{(\ell)} \in (0, 2)$ and $\beta^{(\ell)} \in (-1, 1)$. The hypernetwork is initialized to zero, so $\gamma^{(\ell)} = \mathbf{1}$ and $\beta^{(\ell)} = \mathbf{0}$ at step 0. At initialization the modulated backbone is therefore identical to the unmodulated baseline, giving an exact paired control. A Switch-Transformer-style load-balance term (Fedus et al., 2022) maximizes the entropy of the batch-averaged router distribution, $\mathcal{L}_{\text{regime}} = \lambda_{\text{ent}}(\log R - H[\bar{\alpha}])$, where λ_{ent} is its weight, $H[\cdot]$ the entropy, and $\bar{\alpha}$ the batch-averaged router distribution. This term regularizes marginal regime usage while leaving per-step assignments free to be peaky. We ablate this term away in the strongest-null test, so the observed router uniformity is not attributable to it.

Two application sites. The input-side placement routes the regime into $\Delta_t, \mathbf{B}_t, \mathbf{C}_t$ through the selection mechanism:

$$x_t^{(\ell)} = x_t^{(\ell-1)} + \text{Mamba3}\left(\gamma^{(\ell)} \odot \text{RMSNorm}(x_t^{(\ell-1)}) + \beta^{(\ell)}\right), \quad (\text{input-side})$$

where $x_t^{(\ell)}$ is the output of block ℓ at step t (and $x_t^{(\ell-1)}$ its input, the previous block's output), $\odot$ is the element-wise product, RMSNorm is RMS normalization, and Mamba3($\cdot$) is the selective-scan block above. As a control we also expose a post-scan site that gates the block output, sharing the identical modulator and differing only in where the affine lands:

$$x_t^{(\ell)} = x_t^{(\ell-1)} + \gamma^{(\ell)} \odot \text{Mamba3}\left(\text{RMSNorm}(x_t^{(\ell-1)})\right) + \beta^{(\ell)}. \quad (\text{post-scan})$$

The input-side scale folds into $W_\Delta/W_B/W_C$; the post-scan scale folds into the bias-free output projection W_{out} ; only the post-scan additive shift is non-absorbable (Section 5).

Diagnostics. We gate every claim on activation diagnostics: $\text{film}_g = \text{mean}|\gamma - 1|$ (deviation of the multiplicative gate from identity, zero at identity), its additive counterpart $\text{film}_b = \text{mean}|\beta|$ (which we report for the output-side shift in Section 5), and the router entropy reg_H (whose uniform maximum is $\log R$). A positive generalization gap with an inert FiLM is an artifact, not a regime-conditioning effect.

Family	Objective	Dataset	Gap (sign, unit)	film_g	Verdict
Volatility/scale	Student- t NLL	FI-2010	≈ 0 nats	$\approx \text{ident.}$	NULL
Order-flow volume	Volume NLL	FI-2010	≈ 0 nats	$\approx \text{ident.}$	NULL
Direction	macro-F1	FI-2010	$+0.008^\dagger$ F1	0.018	NULL
Settlement	BCE log-loss	Polymarket	$+0.18^\dagger$ nats	0.008	NULL
PnL	simulated trade	Polymarket	$-\$627^\dagger$	0.010	NULL

Table 1: RegimeFiLM is inert across every objective. reg_H equals its uniform maximum ($\log R$) in all rows. The “Gap” column is in each objective’s own units, so its magnitude is not comparable across rows; only its sign and the inert film_g are. † denotes a non-zero gap whose sign is a distributional artifact, not a FiLM effect. The PnL row’s $-\$627$ is the FiLM A/B gap on the PnL objective (treatment–control), not an economic total. The direction row’s $\text{film}_g = 0.018$ is the endpoint of the *supervised* escalation on that head; the joint-trained value is ≈ 0.004 (Table 3), while the settlement (0.008) and PnL (0.010) rows are joint-trained.

183 4 RegimeFiLM is a decisive null

184 **The result.** Across five objectives (Student- t NLL (volatility/scale), volume NLL¹ (order-
185 flow), direction macro-F1, Polymarket settlement BCE, and a simulated-trading PnL objec-
186 tive) the FiLM parameters collapse to identity under joint training. In every family, on both
187 datasets and on a money objective, film_g decays toward zero while the router entropy reg_H
188 stays at its maximum ($\log R$: $\log 4$ on FI-2010, $\log 8$ on Polymarket). The router is therefore
189 choosing regimes at random rather than learning meaningful assignments. Even when
190 the test metrics look slightly better (direction $+0.008$, settlement $+0.18$, by sign), FiLM is
191 inactive. The treated model is worse in both high and low regimes, and the apparent “gain”
192 is an artifact of how the high–low split and head interact, not real regime conditioning.

193 **Strongest-null escalation.** We force FiLM active (raising the init so $\text{film}_g = 0.150$ at step
194 0) and supervise the router directly on the spot efficiency-ratio axis with the strongest
195 settings (LR multiplier 50, supervision weight 30, decoupled router/FiLM, fed observation
196 volatility). The modulation still decays monotonically toward identity, $\text{film}_g : 0.150 \rightarrow$
197 $0.119 \rightarrow 0.100 \rightarrow 0.074 \rightarrow 0.056 \rightarrow 0.043$, and is still falling at the last step, with $\text{reg}_H = \log 8$
198 throughout. The model has the capacity to use the regime but does not. On FI-2010, a router
199 head trained alone on the volatility bucket reaches 0.92 agreement, a capacity check (a linear
200 probe on the frozen features is weaker, Section 5), yet the FiLM gate on that axis collapses
201 back to identity. On the predictability axis the supervision is weaker (≈ 0.25 – 0.36 agreement),
202 but the same collapse holds there too, reinforcing that the null is optimizer-driven rather
203 than a consequence of missing regime information.

204 **Cross-dataset confirmation and the τ -invariant signature.** We take the finding to two
205 further LOB settings: the Kaggle high-frequency crypto spot book, and FI-2010 re-examined
206 under a predictability-primary regime axis (efficiency-ratio window split, realized volatility
207 secondary). Both are judged by forecasting metrics alone. Across FI-2010 and the Kaggle
208 book (BTC, ETH, ADA at 1 s/1 min/5 min), the zero-initialized modulation never leaves
209 identity: $\text{film}_g \approx 0.004$ and $\text{reg}_H = \log 4$ in every setting (per-asset values in Table 3,
210 appendix). Because FiLM is an exact paired control, every held-out gap is statistically
211 indistinguishable from zero (all 95% CIs include 0). The split is informative, not trivial. The
212 reference windows contain richer structure on all three predictability measures (efficiency
213 ratio, permutation entropy, Hurst), and DeepLOB reaches 0.628 macro-F1 on FI-2010 horizon-
214 10 labels. This shows our null result occurs on genuinely forecastable data rather than being
215 a by-product of a no-signal setting.

216 **The collapse is optimizer-driven, not data-specific.** We run this as a forced-active, ER-
217 supervised escalation, starting at $\text{film}_g = 0.250$. The modulation decays monotonically at

¹The volume-NLL row re-evaluates the Student- t checkpoints on the order-flow channel—a fifth objective, not an independently trained run.

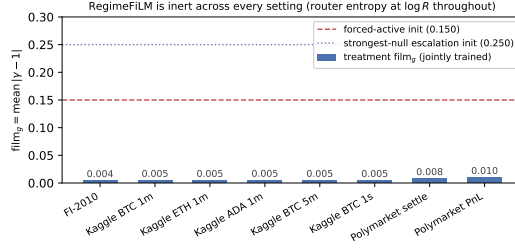


Figure 1: RegimeFiLM is inert across every A/B setting. The jointly-trained treatment $\text{film}_g = \text{mean} |\gamma - 1|$ sits near 0.004–0.010 on every dataset, asset, and resolution, while a forced-active initialization starts at 0.150 and the strongest-null escalation at 0.250. The router entropy sits at its uniform maximum $\log R$ throughout.

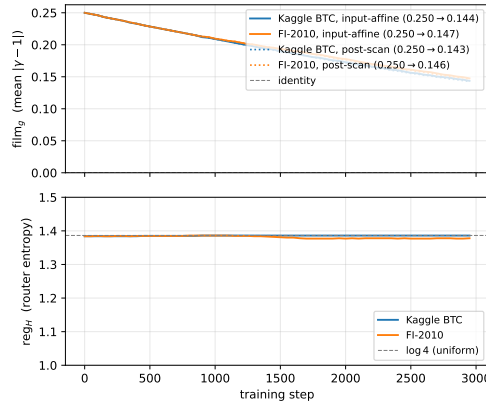


Figure 2: Strongest-null test and its placement-invariance. Forcing the modulation active ($\text{film}_g = 0.250$ at step 0) and supervising the router directly on the efficiency-ratio bucket still decays film_g monotonically toward identity on both datasets, with reg_H pinned at its uniform maximum $\log 4$ throughout. The post-scan gate decays along the same trajectory.

every recorded step on both datasets ($0.250 \rightarrow 0.144$ Kaggle, $0.250 \rightarrow 0.147$ FI-2010), with $\text{reg}_H \approx \log 4$ throughout. The load-balance term is disabled here, so the uniform router is not a load-balance artifact. An exponential fit $ae^{-t/\tau} + c$ matches both trajectories with $R^2 > 0.999$ and near-identical parameters ($a \approx 0.30$, $\tau \approx 7000$ steps): a negative slope sustained to the last logged step, and a fitted asymptote at or below identity ($c \lesssim -0.05$). The modulation extrapolates toward identity, not to a positive plateau. Both markets show almost the same decay rate, which indicates the collapse is driven by the optimizer rather than by any particular dataset (Figure 2). Repeating this escalation on the *best-decoded* volatility bucket instead, the axis a frozen-router head fits to 0.92 on FI-2010, gives the same decay to 0.144 on both datasets ($\tau \approx 6700/6610$, $R^2 = 0.9999$, $\text{reg}_H \approx \log 4$). Supervising the best-encoded axis does not rescue the modulation, so the collapse is gauge absorption, not an undecodable supervised target.

5 Mechanism: an overdetermined null

Why identity is a free direction. The input-side modulation $\tilde{x}_t = \gamma \odot \hat{x}_t + \beta$ is applied immediately before the scan projections $\Delta_t = \sigma(W_\Delta \tilde{x}_t)$, $B_t = W_B \tilde{x}_t$, $C_t = W_C \tilde{x}_t$. Equivalently, substituting the FiLM transformation into the scan gives $\Delta_t = \sigma(W_\Delta(\gamma \odot \hat{x}_t + \beta))$, with analogous expressions for B_t and C_t . Thus the conditioning acts directly through the scan projections themselves. Because these projections are linear, a constant channel-wise affine transformation can be absorbed into their parameters. For any scan projection W ,

$$W\tilde{x}_t = W(\gamma \odot \hat{x}_t + \beta) = W \text{diag}(\gamma) \hat{x}_t + W\beta = W'\hat{x}_t + b', \quad (3)$$

where $W' = W \text{diag}(\gamma)$ and $b' = W\beta$. Thus the scale W' folds exactly into the downstream projection parameters, and the same argument applies to W_Δ , W_B , and W_C . The shift term $b' = W\beta$ can be absorbed only into a trainable bias. In the Mamba3 blocks used here, the Δ -path alone carries one: a trainable dt bias inside the softplus (omitted from the simplified projections above), which absorbs the shift's Δ -component exactly. The **B/C** components are blocked by per-state RMSNorms interposed between the projections and the kernel (a post-norm bias cannot absorb a pre-norm offset), and the gate and scan-input paths carry no bias at all. The shift as a whole is therefore not absorbed; it decays as an unrewarded direction under reconstruction dominance, exactly as the output-side shift does below. Multiple FiLM parameterizations correspond to identical scan dynamics, so the host projections can represent the same constant transformation, which makes identity an optimization-invariant direction. During joint training the projections take over the transformation the modulator previously carried. This lets (γ, β) move toward $(1, 0)$ without materially changing the function the block computes, which is the monotone decay we observe: weight decay drives the modulator toward its identity initialization while the host weights absorb the transformation at negligible loss cost. The descent is therefore driven by regularization on the modulator rather than the task objective, a prediction the weight-decay control below confirms. The absorption is exact only for the constant component of the affine. On a weakly decodable regime the supervised router collapses toward a uniform distribution ($\text{reg}_H \rightarrow \log R$), which makes the modulation increasingly constant. In that limit the conditioning is progressively absorbed into the scan projections, and weight decay drives the remainder to identity.

This is the theory behind the dataset-invariant decay constant. The absorption direction is a property of the block parameterization, not of the data. This does not mean, however, that regime conditioning is impossible in principle. Mechanisms outside this gauge may not be trivially absorbable: modulating $W_\Delta/W_B/W_C$ directly inside the kernel, for instance, or conditioning a quantity inserted between the block's bounding projections. We examine this boundary empirically later in the section.

An output-side placement: the null is overdetermined. The gauge argument names an obvious next test, gating after the scan. We implement it under the identical forced-active, ER-supervised escalation, changing only where the learned (γ, β) are applied. Instead of the input-side $\tilde{x} = \gamma \odot \hat{x} + \beta$, we gate the block output $y' = \gamma \odot y + \beta$. This placement does not escape absorption, just as the criterion predicts. Each Mamba-3 block ends in a bias-free output projection $y = W_{\text{out}}h$, so the output scale folds into it: $\gamma \odot (W_{\text{out}}h) = (\text{diag}(\gamma)W_{\text{out}})h$. This is the same gauge move, now on the block's other face. The shift β is different: with no output bias and a nonlinear RMSNorm before the next block's projections, β is genuinely non-absorbable.

The escalation resolves both at once. Under forced activation and direct ER supervision, film_g and film_b decay monotonically and in lockstep on both datasets ($\text{film}_g : 0.250 \rightarrow 0.146$ FI-2010, $0.250 \rightarrow 0.143$ Kaggle; film_b tracks it within 0.005), with $\text{reg}_H \approx \log 4$ throughout. The exponential fit gives the same dataset-invariant signature as the input-side gate— $a \approx 0.30$, $\tau \approx 7065$ (FI-2010) / 6598 (Kaggle), a negative fitted asymptote ($c \lesssim -0.05$), $R^2 > 0.999$ —so the same monotone decay toward identity holds here too.

The null is therefore overdetermined: no component keeps a surviving direction. Two routes rule them out, and both share a single driver: weight decay acting on a loss that rewards neither. Gauge absorption removes every direction the host weights can represent, namely the multiplicative scale on either face of the block. This is why the decay constant is invariant to placement. Reconstruction dominance removes the one direction the host weights cannot, the non-absorbable shift β . Because β earns nothing under the objective, it returns to identity at the same rate, decaying in lockstep with γ . Escaping both would require a modulation that acts between the block's bounding projections: one that modulates $W_\Delta/W_B/W_C$ within the fused selective-scan kernel. That is the kernel-level change beyond our hardware envelope.

The boundary, demonstrated. We make this concrete at toy scale on the pure-PyTorch reference scan (the slow path `mamba.ssm` ships, which the fused-kernel hardware limit does not touch). An input-affine modulation folds into the host projections to floating-point precision (max output deviation 1.2×10^{-6}) and decays to identity under joint training with weight decay ($\text{film}_g : 0.086 \rightarrow 0.010$). The same modulation applied to the projected $\Delta/\mathbf{B}/\mathbf{C}$ inside the scan is provably non-absorbable (best-fit fold residual 1.3×10^{-2} , $\sim 10^4 \times$ larger) and persists ($\text{film}_g : 0.093 \rightarrow 0.175$, seeds 0–3). A weight-decay control isolates the driver. With weight decay removed, the full-model escalation no longer decays: the time constant blows up from $\tau \approx 6685$ ($R^2 = 0.9999$) to $\tau \approx 3 \times 10^5$ ($R^2 = 0.48$, no clean decay), and the toy reproduces the same flattening (input-affine $0.086 \rightarrow 0.342$ with weight decay off vs. $0.086 \rightarrow 0.010$ with it on). The gauge-direction decay is therefore driven by weight decay over a flat loss, not by the objective.

The regime is decodable yet unused. Two probes localize the failure. The jointly-supervised router is uniform per step (per-sample entropy $\approx \log 4$, near-chance ER agreement 0.25–0.31): supervision does not even make it regime-discriminative. A frozen linear probe decodes the predictability bucket only weakly (0.33–0.36 vs. 0.25) and volatility better (0.43–0.47)—yet the volatility-axis gap is also null. So even on the best-encoded axis, and even with the regime supplied by supervision, the in-scan modulation collapses to identity.

Language-model control: the null is bounded by regime decodability. Because the collapse is optimizer-driven, the natural worry is that it is nonetheless specific to LOB data. A minimal language-model control maps where the boundary lies. We train a 2-layer char-level Mamba LM (built directly on `mamba.ssm` blocks) on a two-domain corpus of English prose versus Python source. A context-aware router, supervised on the domain, infers the per-position regime, and the same input-affine FiLM gates the second block. Here the regime is strongly decodable: the router reaches ≥ 0.9 domain accuracy (0.98 at peak, 0.91 at the last logged step), and the FiLM does not collapse. Instead film_g rises from 0.11 to 0.52 and plateaus, the opposite of the LOB decay. This sharpens the mechanism rather than counters it. Gauge absorption removes the constant component of the affine unconditionally, while the input-dependent component survives only when the regime is decodable enough to be worth encoding. We therefore scope the claim accordingly: the optimizer-driven collapse holds for weakly-decodable regime conditioning of selective scans (the predictability and volatility axes a financial world model actually faces), not for all conditioning. This is a single-seed toy; the router-to-persistence outcome is clear, but the plateau value is not load-bearing.

The criterion, distilled. The mechanism reduces to one structural test.

Proposition (absorbability criterion). Let a channel-wise affine (γ, β) act on an activation immediately consumed by a learned linear map A (optional bias b). The *scale* is always gauge, $A(\gamma \odot x) = (A \text{diag}(\gamma))x$, absorbed into A with or without a bias; the *shift* is gauge only when A carries a trainable bias and no nonlinearity intervenes, $A(x + \beta) + b = Ax + (b + A\beta)$. A mechanism that instead modifies the in-kernel scan parameters W_Δ, W_B, W_C —or any quantity strictly between a block’s bounding projections—is *not* gauge-absorbable. *Corollary:* in the Mamba3 configuration used here, input- and output-side FiLM share the same split: the scale is gauge ($W_{\{\Delta, B, C\}} \text{diag}(\gamma)$ or $\text{diag}(\gamma)W_{\text{out}}$), while the shift is non-absorbable as a whole (only its Δ -component can fold, into the trainable dt bias) and decays because reconstruction dominance removes any direction the loss does not reward.

6 Demonstration substrate and architecture-agnosticism

The mechanism above is the subject. The financial machinery is the substrate that let us probe it on the three genuinely different distributions of Section 3. We compress it here to what the null and its agnosticism check require.

Substrate. We build a Mamba-3 world model for LOB microstructure in the DreamerV3/Drama lineage: a Transformer-over-depth encoder, a stack of Mamba-3 SISO blocks,

Backbone	Params	FI-2010 recon NLL	Kaggle recon NLL
Mamba-1	11.45M	−0.5403	0.0241
Mamba-2	10.20M	−0.5309	−0.0213
Mamba-3 SISO	10.30M	−0.5346	0.0207
Transformer (post-norm)	9.90M	−0.5175	0.0191
Transformer (pre-norm)	9.61M	−0.5041	0.0224

Table 2: Matched-parameter backbone ablation (held-out reconstruction NLL, lower is better). No backbone wins both datasets—Mamba-1 leads on FI-2010, Mamba-2 on Kaggle—so the choice of sequence model is not the operative variable.

and a categorical-latent dynamics prior under a reconstruction objective. Of these, only the settlement and book-relative-EV heads bear on any result below. For Polymarket we expose the spot path that defines settlement (Chainlink close vs. open), plus a time-to-expiry clock and TTE-weighted target.

Backbone ablation. Under matched budgets ($\sim 10.3\text{M}$, Table 2), seed 0, 3000 steps on FI-2010 and the Kaggle BTC book, the held-out Student- t reconstruction NLL is a genuine ambiguity. No backbone leads on both datasets, and a pre-norm Transformer matched to the Mamba-3 block is competitive but non-dominant, not an under-tuned-baseline artifact. This is exactly what a result about how selective scans train, rather than which one is best, should predict.

We run Mamba-3 SISO throughout. The MIMO kernel exceeds our GPU’s shared-memory cap at every warp-valid (rank, chunk) configuration, which is the same kernel-level region the null’s escape route would occupy.

The economic coda is architecture-independent. Given the spot path, the model emits a directional settlement probability, traded only when a train-frozen efficiency-ratio gate fires. We summarize it with a market-block bootstrap (whole markets resampled, so correlated trades count once). Where the gate targets, this is a survivable edge ($P(\text{profit}) \approx 1.0$, 95% CI $[\text{+\$2,445}, \text{+\$7,747}]$ over its 127 traded markets). The edge is architecture-independent. On the identical 645-market held-out set, a logistic regression on the same spot-conditioned features matches and exceeds it ($\text{+\$6,914}$ vs. the settlement head’s $\text{+\$5,147}$), because binary settlement is near-deterministic in the spot move since open. The operative variable is therefore the features and the gate, not the backbone. Depth-band cuts, the tradeable-EV head, and the reliability diagnostic are deferred to the long-form version.

Limitations. The null is multi-setting and multi-seed; the constructive claims that bound it are smaller. The optimizer attribution rests on a τ matched across two datasets plus a within-model weight-decay ablation. The escalation fits are seed-0, with the seeds-0–3 in-kernel toy supplying the multi-seed check. The language-model control and the in-kernel “survives” result are single-seed and at toy scale. The substrate is finance-only, so the gauge argument’s architecture-universality is algebraic, not demonstrated at FM scale. None of this bears on the null itself.

7 Conclusion

The most natural, kernel-friendly way to make a Mamba backbone regime-aware is an input-affine FiLM on each block. It is a decisive training-dynamics null: the modulation collapses to identity under joint training and cannot be forced active even by direct supervision. The cause is a gauge symmetry. The affine folds losslessly into the block’s own $W_\Delta/W_B/W_C$ projections, so the optimizer slides it back to identity ($\tau \approx 7000$, $R^2 > 0.999$), and an output-side placement makes the null overdetermined. The upshot is architecture-general by the algebra of the fold: only a placement between a block’s bounding projections, inside the fused kernel, can persist.

Use of Large Language Models

In keeping with the COLM 2026 policy on disclosure of LLM use, we report that a large language model was used while preparing this paper. It assisted in drafting and editing the prose, including the phrasing of the reference list, and was used to assist with evaluation.

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A Cross-dataset detail

Dataset	Asset / resolution	Gap (pred / vol)	film_g	Verdict
FI-2010	event stream	$\approx 0 / \approx 0$	0.004	NULL
Kaggle BTC	1 min	$\approx 0 / \approx 0$	0.004	NULL
Kaggle ETH	1 min	≈ 0	0.005	NULL
Kaggle ADA	1 min	≈ 0	0.005	NULL
Kaggle BTC	5 min	≈ 0	0.005	NULL
Kaggle BTC	1 s (slice)	≈ 0	0.005	NULL

Table 3: Cross-dataset confirmation of the regime-FiLM null (per-asset breakdown of the summary in Section 4). $\text{reg}_H = \log 4$ (uniform router) in every row. The null is dataset-, asset-, and resolution-independent.